

UDDIPA DATTUDU CHAVALI

India | +91-9392741655 | chavaliuddipadattudu@gmail.com | [My Portfolio Website](#)
linkedin.com/in/chavaliuddipadattudu | github.com/Chavali-Uddipa-Dattudu

ACADEMIC & RESEARCH PROFILE

Electronics and Communication Engineering undergraduate focused on VLSI, RF CMOS IC design, and analog circuit design, with hands-on research experience in RFIC and CMOS image-sensor circuits. Experienced in transistor-level circuit design, simulation, and analysis using Cadence Virtuoso, SPICE, and Verilog, with additional experience in Python-based EDA automation. Interested in developing practical, high-performance integrated-circuit solutions across analog, RF, and mixed-signal domains.

ACADEMIC BACKGROUND

Madanapalle Institute of Technology and Science <i>Bachelor of Technology in Electronics and Communication Engineering; CGPA: 9.37</i>	Andhra Pradesh, India July 2025 – Expected May 2028
ESC Government Polytechnic College <i>Diploma in Electronics and Communication Engineering; Percentage: 95.15%</i>	Andhra Pradesh, India Aug 2022 – May 2025

TECHNICAL SKILLS

Core Competencies: Analog Circuit Design, Radio Frequency CMOS Integrated Circuit Design, DC/AC/Transient Simulations.
EDA Tools: Cadence Virtuoso, Altium Designer (Basic PCB), NGSPICE, NI Multisim, MATLAB.
Programming & HDLs: Verilog HDL, Python (Currently learning Data Science & Basic Automation), C language.

ACADEMIC, RESEARCH AND INDUSTRIAL EXPERIENCE

National Institute of Technology (NIT) Calicut <i>Intern - Analog IC Design for Image Sensors</i>	Remote July 2026 – Present
- Collaborating on remote research under the supervision of Dr. Bhuvan B, focusing on the study and design of Programmable Gain Amplifiers (PGAs) and Single-Slope ADCs for CMOS camera image sensors. - Working on the transistor-level modeling and simulation of the converter blocks to optimize gain, noise, and power for high-resolution imaging applications.	
National Institute of Technology (NIT) Calicut <i>Summer Research Intern - RFIC Design for Anti-Drone Radar</i>	Kerala, India May 2026 – July 2026
- Designed and simulated an unconditionally stable 9.0 GHz cascode Low Noise Amplifier (LNA) in a 180nm CMOS process using Cadence Virtuoso under the guidance of Dr. Karthik R, achieving a forward voltage gain (S_{21}) of 11.06 dB and a Noise Figure (NF) of 1.41 dB. - Engineered an active LC L-match output network and multi-finger gate topologies (30 fingers of $5.0\ \mu\text{m}$) to neutralize the Miller effect, ensuring precision $50\ \Omega$ impedance matching ($S_{11} = -17.06\ \text{dB}$, $S_{22} = -23.27\ \text{dB}$). - Verified high power efficiency via ADE-L DC operating point extraction, yielding a low power consumption (P_{DC}) of 14.13 mW and an exceptional Figure of Merit (FoM) of 21.05 GHz/mW for X-band receiver front-ends. - Actively collaborated with the research group, presenting weekly simulation data, design trade-offs, and performance metrics to the principal investigator to iteratively optimize the RF front-end architecture.	
Southern Power Distribution Company of A.P. Limited (APSPDCL) <i>Network Intern in TELECOM Department (Industrial Training as part of my Diploma)</i>	Andhra Pradesh, India May 2024 – Nov 2024
- Completed a 6-month industrial training focused on telecommunications and the physical layer of networking. - Configured and troubleshooted enterprise hardware, including Cisco routers, SD-WAN setups, and 24-port HP/D-Link switches to keep power grid data flowing. - Gained practical experience with physical infrastructure by crimping Cat6 cables, punching down keystone jacks, and observing the troubleshooting of fiber optic (OFC) links.	

ENGINEERING PROJECTS

Dual-Input LiPo Battery Charger PCB <i>Altium Designer</i>	Aug 2026
- Designed a custom 2-layer PCB for a lithium-ion battery charger, including dual power inputs (Micro-USB and DC barrel jack) safely separated by input OR-ing diodes. - Used a bottom copper pour to connect all grounds and keep the board cool. - Added a physical solder jumper to let users switch between 100mA and 500mA charging speeds, mapped 3D component bodies to check clearances, and printed clear silkscreen safety warnings.	

Automated CMOS Inverter Characterization | *Python, NGSPICE, Pandas*

July 2026 – Aug 2026

- Wrote a Python automation script that runs NGSPICE silently in the background, automatically sweeping 180nm CMOS transistor widths from 1.0 μm to 5.0 μm . Used Pandas to extract and parse the raw simulation data directly from text files, replacing manual GUI testing to efficiently analyze Voltage Transfer Characteristics (VTC).
- Graphed the shifting switching thresholds using Matplotlib to visually prove how changing the PMOS size alters the strength of the pull-up network.

180nm 6T SRAM Cell Design and Characterization | *Analog VLSI Design*

July 2026

- Designed a 6T SRAM cell in a 180nm CMOS process using Cadence Virtuoso, optimizing transistor sizing (Cell Ratio = 2.0, Pull-Up Ratio = 0.7) for balanced read/write stability.
- Extracted Voltage Transfer Characteristics (VTC) and applied Seevinck's method via a custom MATLAB script to calculate a robust Read Static Noise Margin (SNM) of 309.0 mV.
- Performed transient and DC analyses to validate energy efficiency, achieving an active dynamic power of 86.39 μW (at a 40ns period) and an ultra-low static leakage of 24.1 pW.

RELEVANT COURSEWORK

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|---|-----------------------------------|---------------------------------|
| • Analog Circuits | • Digital System Design (Verilog) | • Electronic Devices & Circuits |
| • Network Theory | • Python Programming | • PCB Design |
| • Introduction to Quantum Technologies and Applications | | |

CERTIFICATIONS

Microsensors and Nanosensors (In Progress) | *NPTEL / IIT Guwahati*

July 2026 – Oct 2026

System Design Through Verilog | *NPTEL / IIT Guwahati*

July 2026 – Sept 2026

- Studied advanced digital design concepts including behavioral modeling, data flow modeling, and the synthesis of combinational and sequential logic.

VLSI for Beginners | *NIELIT Calicut (MeitY C2S Project)*

Aug 2026

Introduction on Intellectual Property to Engineers | *NPTEL / IIT Kharagpur*

Jan 2026 – Mar 2026

- Gained essential knowledge of patent filing procedures, trade secrets, and IC Layout Design protection frameworks.

Understanding Incubation and Entrepreneurship | *NPTEL / IIT Bombay*

Jan 2026 – Apr 2026

- Studied technology-led entrepreneurship, customer discovery processes, and the Lean Canvas Model to understand the framework for converting technical solutions into viable businesses.

WORKSHOPS & INDEPENDENT STUDY

Radio-Frequency Integrated Circuits (Independent Audit) | *NPTEL / IIT Kanpur*

July 2026 – Present

- Auditing coursework focused on wireless receiver/transmitter architectures, studying RF metrics, LNA design, mixers, and Power Amplifier (PA) architectures.

Seminar: VLSI Testing and Physical Design | *MITS ISTE Student Chapter*

Feb 2026

Seminar: VLSI and Electromagnetic Industry Practices | *MITS IETE Students Forum*

Feb 2026

Guest Talk: India's Semiconductor Revolution & VLSI | *MITS Alumni Association*

Aug 2025

HONORS & AWARDS

1st Prize (INR 25,000), Innovation Nexus 2026 Hackathon | *MITS, PathVision & IITNiF*

Aug 2026

- Awarded 1st place out of 50 competing teams during a rigorous 16-hour hackathon for solving a targeted industry problem statement to develop a "Real-Time Agricultural Transport Condition Monitoring System".
- Led project research, feasibility analysis, and architecture study, ultimately pitching the solution to the jury.
- Programmed an ESP32 microcontroller and physically integrated multi-modal sensors (DHT11, MQ-4, GPS) to build the IoT data ingestion pipeline.

National Level Technical Symposium Wins | *Multiple Institutions*

2024 – 2026

- **2nd Position, "Circuit Detective":** ECLECTICA 2K26 at MITS, Madanapalle (April 2026). Awarded a cash prize.
- **Double 1st Prize, "Circuit Hunt" & "Technical Quiz":** E-Merge 2K24 at JNTUACEA (Nov 2024). Awarded a total of INR 2000 cash prize for winning both events.
- **1st Prize, Paper Presentation:** Ripple National Symposium at RGM College of Engineering, Nandyal. Awarded an INR 1000 cash prize.

Consistent Academic Excellence & Campus Awards | *ESC Govt Polytechnic College*

2023 – 2025

- Consistently ranked at the top of the cohort across all semesters (2023–2025), receiving multiple faculty-awarded merits and cash prizes for highest examination results.
- Secured 1st and 2nd place finishes in consecutive campus-wide technical quizzes (2023).